Semi & anti join, subqueries

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Semi join
Chooses records in the first table where the condition is met in the second table

Example:
SELECT president, country, continent
FROM presidents
WHERE country IN
    (SELECT country
    FROM states
    WHERE indep_year < 1800);

Anti join
Chooses records in the first table where col1 does NOT find a match in col2

Example:
SELECT country, president
FROM presidents
WHERE continent LIKE '%America'
    AND country NOT IN
        (SELECT country
        FROM states
        WHERE indep_year < 1800);

Subqueries in WHERE
Useful for intermediary transformations
Can be in any part of a query
It can return scalars, lists, tables
Use cases: comparing groups to summarized values, reshaping data, combine unjoinable data

```
SELECT home_goal
FROM match
WHERE home_goal > (
    SELECT AVG(home_goal)
    FROM match);
SELECT AVG(home_goal) FROM match;
```

It requires an alias!

Subqueries in FROM
Restructure and transform data
Calculating aggregates of aggregates
I can create multiple subqueries in one FROM statement-alias and join them!
I can join a table to a subquery in FROM if the subquery has a column I can use for the join

Example:
SELECT DISTINCT *(remove duplicates)* monarchs.continent, sub.most_recent
FROM monarchs,

```
        (SELECT continent, MAX(indep_year) AS most_recent
        FROM states
        GROUP BY continent) AS sub
WHERE monarchs.continent = sub.continent
ORDER BY continent;
```

Subqueries in SELECT

Returns a **single value**-include aggregate values to compare to individual values
It will generate an error if the output is not a single value
Make sure to have filters in the correct places! Subquery is processed before main, so filter it if needed
Used in mathematical calculations (deviation from the average)

Example:

```
SELECT DISTINCT continent,
        (SELECT COUNT(*)
        FROM monarchs
        WHERE states.continent = monarch.continent) AS monarch_count
FROM states;
```

Subqueries in SELECT

```
SELECT
  date,
  (home_goal + away_goal) AS goals,
  (home_goal + away_goal) -
     (SELECT AVG(home_goal + away_goal)
      FROM match
      WHERE season = '2011/2012') AS diff
FROM match
WHERE season = '2011/2012';
```

```
| date       | goals | diff               |
|------------|-------|--------------------|
| 2011-07-29 | 3     | 0.28354037267081   |
| 2011-07-30 | 2     | -0.71645962732919 |
| 2011-07-30 | 4     | 1.28354037267081   |
| 2011-07-30 | 1     | -1.71645962732919 |
```